

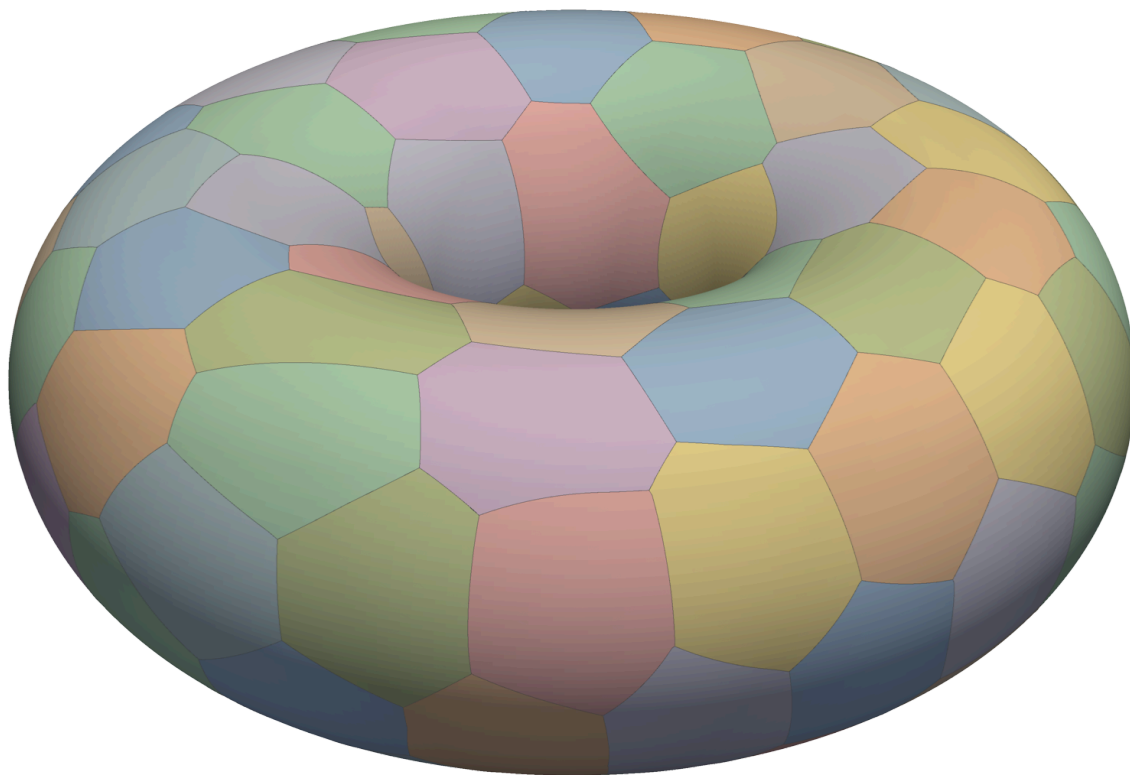
Minimal-perimeter partitions on surfaces

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Status brief

The relaxation pipeline, its limits, and the fix that reaches 300 cells

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$n = 100$ cells on a torus with 114,144 mesh vertices, after the contour optimisation.

- A surface $S \subset \mathbb{R}^3$ and a partition $(\omega_i)_{i=1}^n$ of S into n cells of prescribed area $|\omega_i| = c_i = |S|/n$.
- Minimise the total geodesic perimeter (Bogosel & Oudet):

$$\text{Per}(\omega_1) + \cdots + \text{Per}(\omega_n).$$

- Working surface: the torus with $R = 1, r = 0.6$, triangulated at up to 114,144 vertices.
- Two phases, as in the paper: a **relaxed** problem on densities, solved on a ladder of finer meshes; then the extracted **contours** optimised directly under exact area constraints.

The relaxed problem

$$F_\varepsilon(\mathbf{u}) = \sum_{i=1}^n \int_S \left(\varepsilon |\nabla_\tau u_i|^2 + \frac{1}{\varepsilon} u_i^2 (1 - u_i)^2 \right) d\sigma, \quad \mathbf{u} \in X = \left\{ \int_S u_i = c_i, \sum_{i=1}^n u_i = 1 \right\}$$

- One density $u_i : S \rightarrow [0, 1]$ per cell. **Gradient term:** an interface costs $\varepsilon |\nabla_\tau u_i|^2$, so transitions spread over a width $\sim \varepsilon$ and short interfaces are cheap. **Potential term:** the double well $u_i^2 (1 - u_i)^2$ is zero only at 0 and 1, so away from interfaces u_i is pushed to $\{0, 1\}$.
- **Γ -convergence** (Theorem 2.2): as $\varepsilon \rightarrow 0$, $F_\varepsilon \xrightarrow{\Gamma} \frac{1}{3} \sum_i \text{Per}(\{u_i = 1\})$, so minimisers of F_ε on X approach minimal-perimeter partitions with the prescribed areas.
- ε must be of the order of the mesh size – smaller and the gradient term sees nothing. Hence the **mesh ladder**: decrease ε with each refinement, interpolating the previous densities as the starting point.

Constraints

X carries the two constraints the partition needs: **equal areas**

$\int_S u_i = |S|/n$ and the **partition of unity** $\sum_i u_i = 1$ at every point.

Both are imposed exactly, by orthogonal projection after every gradient step.

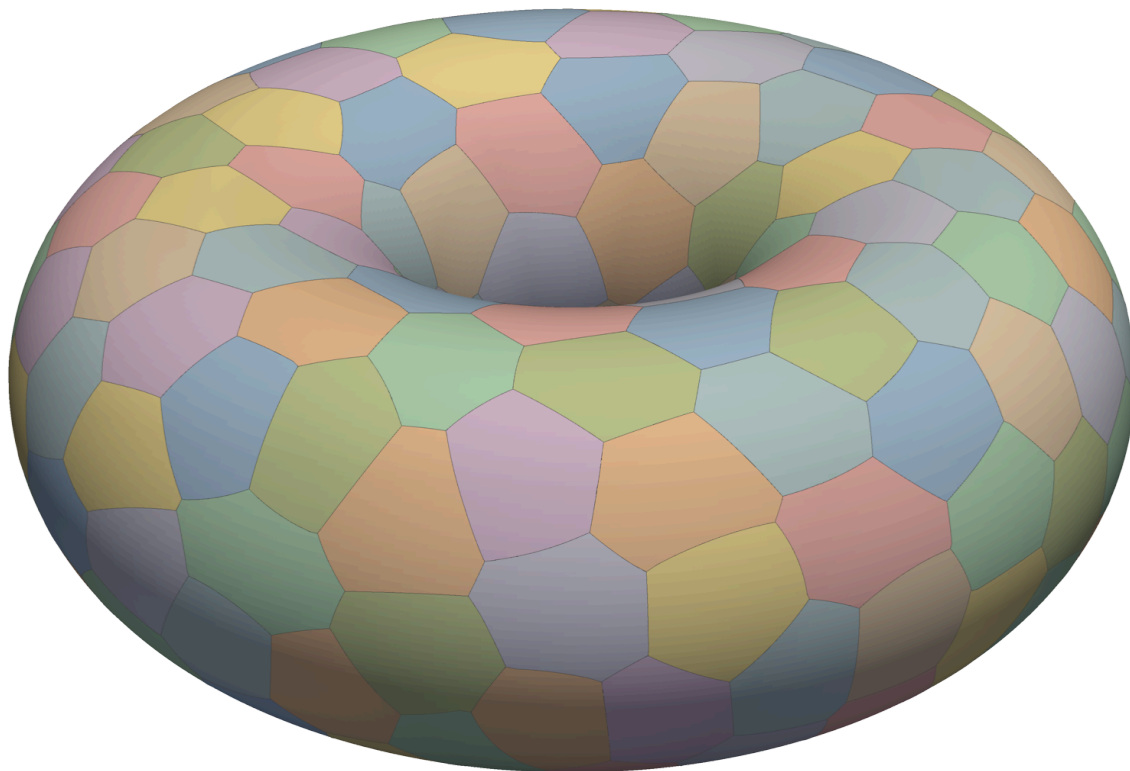
Finite elements: the three terms and their gradients

Mesh with N vertices; P_1 mass and stiffness matrices M, K ; nodal values $\varphi^i \in \mathbb{R}^N$ of u_i ; lumped mass $v = \mathbf{1}^\top M$, so $\langle v, \varphi^i \rangle \approx \int_S u_i$; $A = |S|$; $\odot$ = component-wise product.

term	discrete energy, per cell i	gradient w.r.t. φ^i	role
gradient	$\varepsilon (\varphi^i)^\top K \varphi^i$	$2\varepsilon K \varphi^i$	short interfaces
potential	$\frac{1}{\varepsilon} (w^i)^\top M w^i, \quad w^i = \varphi^i \odot (1 - \varphi^i)$	$\frac{2}{\varepsilon} (1 - 2\varphi^i) \odot M w^i$	crisp 0/1 values
penalty	$\lambda \left(1 - \text{Var}_v(\varphi^i)/T\right)$	$-\frac{2\lambda}{AT} v \odot (\varphi^i - \mu)$	penalises a diffuse φ^i

$\text{Var}_v(\varphi^i) = \frac{1}{A} \sum_k v_k (\varphi_k^i - \mu)^2$ with $\mu = \langle v, \varphi^i \rangle / A = 1/n$, and $T = \frac{1}{n} \left(1 - \frac{1}{n}\right)$ is the variance of a sharp indicator of area A/n – so the penalty vanishes for a crisp cell: the paper's $\lambda (\text{std}(\varphi^i) - s_{\text{target}})^2$, normalised. The energy is the sum of the three rows over $i = 1, \dots, n$.

- **Constraints on $\Phi = (\varphi^1 \dots \varphi^n) \in \mathbb{R}^{N \times n}$:** every row sums to 1 (partition of unity), $\langle v, \varphi^i \rangle = A/n$ for every column (equal areas); orthogonal projection onto both after each gradient step (the paper's Algorithm 1). $\varepsilon = \sqrt{\text{mean triangle area}} \approx h$.
- The paper's text has $w = \varphi^2 \odot (1 - \varphi)^2$, i.e. $\int u^4(1 - u)^4$; we use $w = \varphi \odot (1 - \varphi)$, which is the $\int u^2(1 - u)^2$ of F_ε .



$n = 200$ on the same mesh, after contour optimisation. Lines: the optimised contours.

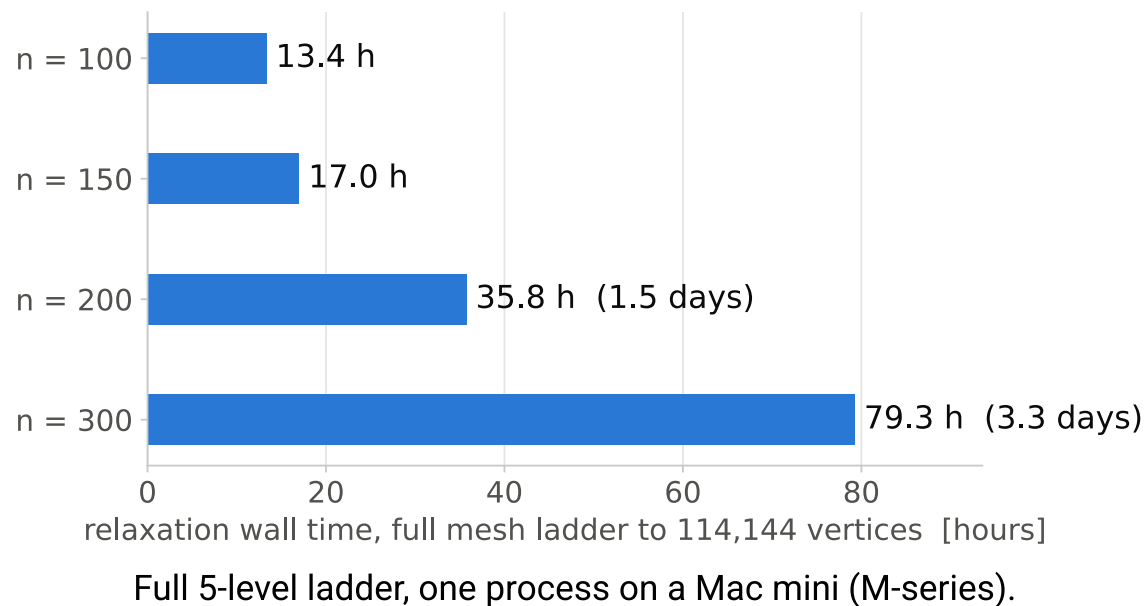
- **Readout (paper's 5.1).** $\phi_i(x) = 1$ if $u_i(x) \geq \max_{j \neq i} u_j(x)$, else 0: every vertex goes to the cell with the largest density. That hard labelling is the partition.
- **Contours.** The 0.5 level sets of the ϕ_i cross mesh edges at *variable points* $x_i = \lambda_i v_1 + (1 - \lambda_i) v_2$, $\lambda_i \in [0, 1]$; the void at each triple point is closed by a Steiner tree through the Fermat point.
- **Optimisation.** Minimise $\sum_i \text{Per}(\omega_i)$ over the λ_i with exact equal-area constraints (interior point, analytical derivatives). When a λ_i reaches 0 or 1, or a Fermat point leaves its triangle, the contour is rebuilt and the optimisation restarts.
- **Three validity checks on the readout, before the contour stage:** no empty cell; every cell within $\pm 5\%$ of equal area; every cell connected.

n	λ	relaxation wall	empty	imbalanced	fragmented	worst cell	final perimeter
100	5.1	13.4 h	0	0	0	0.78 %	185.25
150	6.0	17.0 h	0	0	0	1.24 %	228.16
200	11.0	35.8 h	0	0	0	1.65 %	262.11

All three checks pass on the raw readout. "Worst cell" is the largest deviation from equal area *before* the contour stage, which then equalises the areas to solver tolerance.

What it took — the practical problems with the relaxation:

- **Seeded initialisation is mandatory.** From random densities the energy traps in the symmetric diffuse state ($n = 30$: 43 % worst-cell error vs 0.7 % seeded). We start from Voronoi cells of well-spread seed vertices.
- **λ has a working window that moves with n** (≈ 5 at $n = 100$, ≈ 11 at $n = 200$). Too low: diffuse "runt" cells. Too high: the penalty dominates, levels stop early, interfaces never crisp.
- **Some failures are seed-specific.** $n = 200$ was unblocked by changing the seed. A run is one trajectory, not a statistic.



- Grows much faster than n: $\times 2.7$ from $n = 100$ to 200 , $\times 2.2$ again to 300 .
- **93 % of the relaxation is the projection** onto the two constraints — an alternating projection, ~ 46 inner iterations per gradient step at $n = 300$.
- Part is waste: at $n = 100$ the coarsest level runs to its 30,000-iteration cap with labels frozen from iteration $\sim 6,000$ (32 % of that ladder). A structure-based stop reclaims it (13.4 h $\rightarrow$ 9.7 h) but does not change the trend.
- The contour stage is 30–40 min at these n . The relaxation is the cost.

What goes wrong as n grows

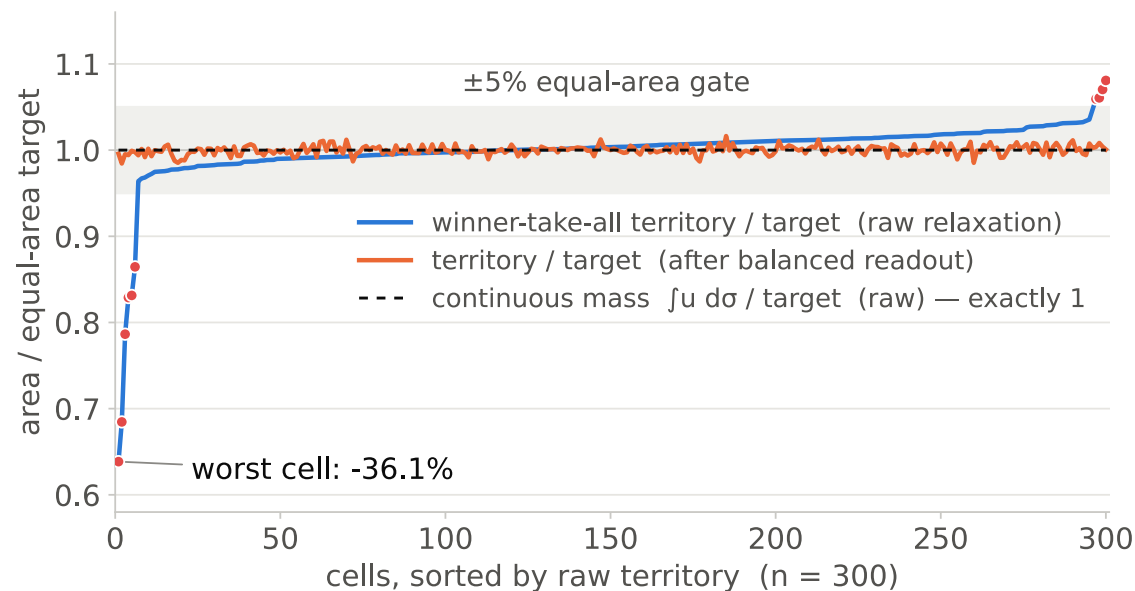
Three artefacts of the winner-take-all readout, and why

Three artefacts, one cause

	Empty cell	Runt cell	Split cell
territory after readout	zero — wins no vertex	well below target (−36 % at $n = 300$)	right total, in 2+ disconnected islands
peak density	low (~ 0.1), never a winner	1.0 — a confident winner on a small core	1.0 in every piece
what breaks	wrong cell count	unequal areas → contour stage infeasible at iteration 0	non-physical topology → multi-loop contours
caught by	empty-cell check	area check	connectivity check only
status	resolved — seeded init	resolved to $n = 200$ (corrected well + moderate λ); back at $n = 300$	from $n \approx 200$ (3/200 on one seed, 2/300); rare, never zero

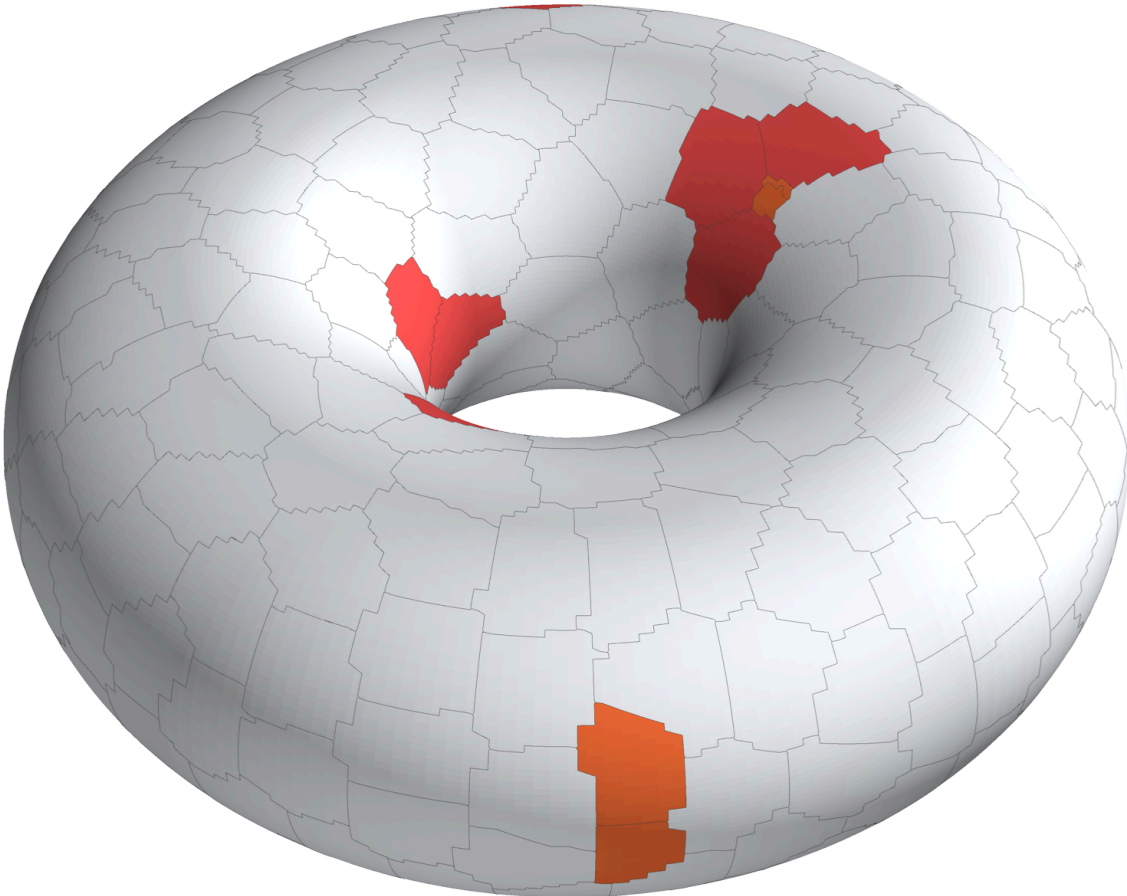
Each is a perfectly consistent **minimiser of F_ϵ** — and not a valid n -cell partition. The checks catch them; nothing in the relaxation prevents them.

Why: the energy controls mass, the readout takes territory



$n = 300$ on a 47,488-vertex mesh: raw relaxation and its balanced readout.

- The area constraint fixes the **continuous mass** $\int_S u_i = |S|/n$ exactly — "equal paint". The readout (5.1) awards **territory** by the maximum. Nothing in F_ε or in X refers to that maximum.
- A healthy cell — $u_i \approx 1$ on a compact blob — has the two agree. A cell that spreads its paint into the diffuse band between cells keeps its mass and **loses territory**: the runt.
- Nothing penalises a cell in two pieces, and equal mass is **non-local**: a cell can buy area far from its core. Fragmentation is born mid-level, while the imbalance is being reduced.
- The levers that work at moderate n — seeded init, the corrected well, λ — run out at $n = 300$: λ has a ceiling above which interfaces never crisp.

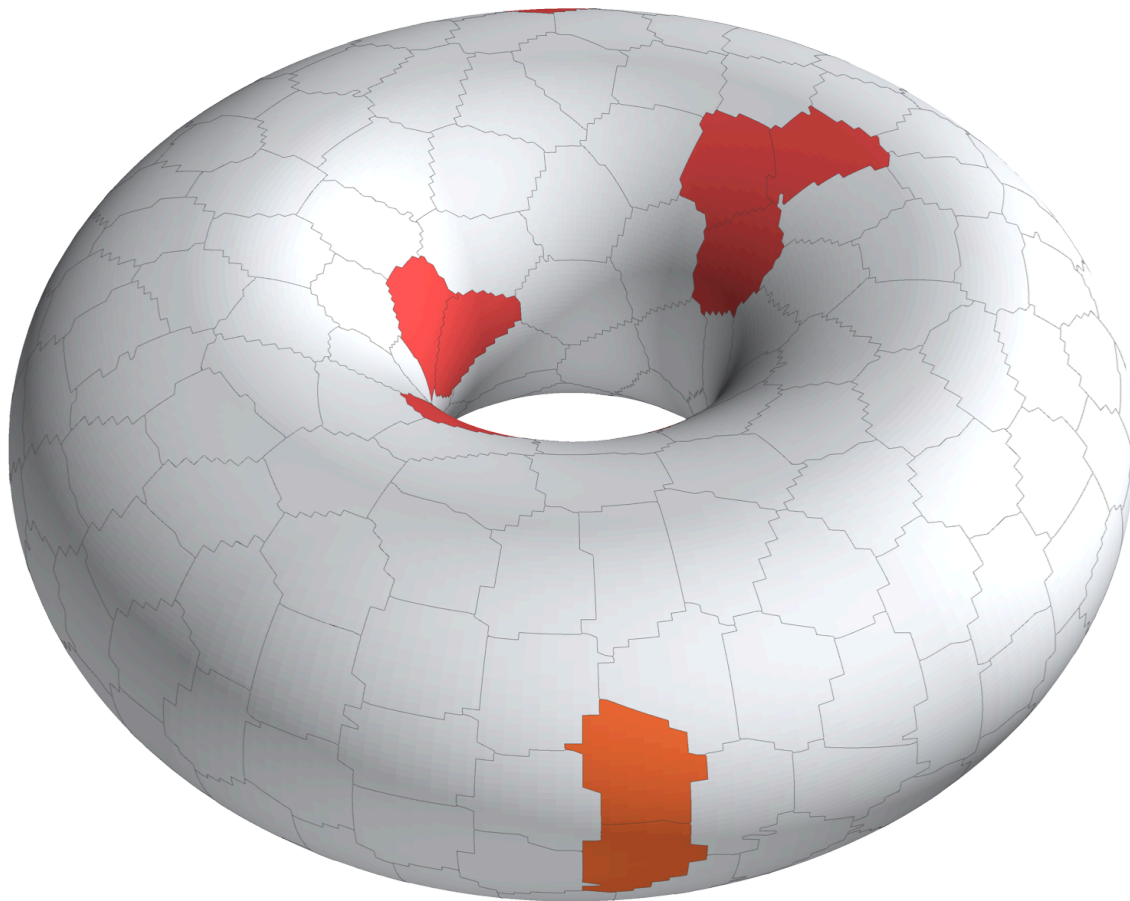


Raw readout, 47,488 vertices, before the contour stage. Red: outside the $\pm 5\%$ area check; orange: fragmented (one stray piece sits beside the red cells).

mesh vertices	relaxation	empty	imbalanced (worst)	split
47,488	22.0 h	0	10 (-36.1 %)	2
114,144	+57.3 h	0	3 (-24.8 %)	2

- Same $\lambda = 11.5$ and seed. The finer mesh **reduces** the imbalance, does not remove it; the split cells survive it.
- The contour stage cannot start here: the worst cell's deficit *is* the equal-area violation at iteration 0, and the optimiser stalls infeasible.
- The two coarse levels did no work (32 and 83 vertices per cell – the line search floors within ~ 60 iterations); level 2 did all of it.

The fix: a balanced readout, at extraction time



The same cells after the balanced readout: all within $\pm 1.63\%$ of equal area, all connected.

- Leave F_ε alone; fix the **readout**. Replace $\arg \max_i u_i$ by $\arg \max_i [\log u_i + \psi_i]$: n per-cell offsets solved so every cell's territory hits $|S|/n$ – the dual of an optimal-transport assignment. Raising ψ_i grows cell i from its own core, so the correction is **local**.
- Then **connectivity repair**: a stray piece joins the neighbour with the longest shared boundary; equal areas are restored by single-vertex transfers that never disconnect a cell.
- **$n = 300$, 47k vertices**: 10 imbalanced / -36.1% / 2 fragmented $\rightarrow$ offsets: 0 / 2 $\rightarrow$ repair: **0 / 1.63 % / 0**, in ~ 17 s. 1,840 vertices relabelled, boundary $+1.9\%$.
- Same file format, so the contour stage runs unchanged: **323.32** (47k), **322.96** (114k).

n	mesh vertices	route	worst cell	final perimeter
100	114,144	raw relaxation	0.78 %	185.25
150	114,144	raw relaxation	1.24 %	228.16
200	114,144	raw relaxation	1.65 %	262.11
300	114,144	relaxation + balanced readout	0.64 %	322.96

- **Valid, equal-area, connected partitions for $n = 100, 150, 200$ and 300 on one common mesh**, exported for downstream use.
- The balanced readout closes the winner-take-all gap **at any n , in seconds** — that part is solved.
- **The relaxation itself is the open problem:** 79 h at $n = 300$, rising faster than linearly, and its raw partition already needs repair. Going past 300 cells means changing how the relaxation arrives at a labelling — that is the next conversation.